Bucket Sort

Assumption:

the input is drawn from a uniform distribution in [0,1)

Performance:

average-case running time of O(n)

Buckets:

divides the interval into n equal-sized subintervals

Expectation:

numbers in the same bucket will be limited

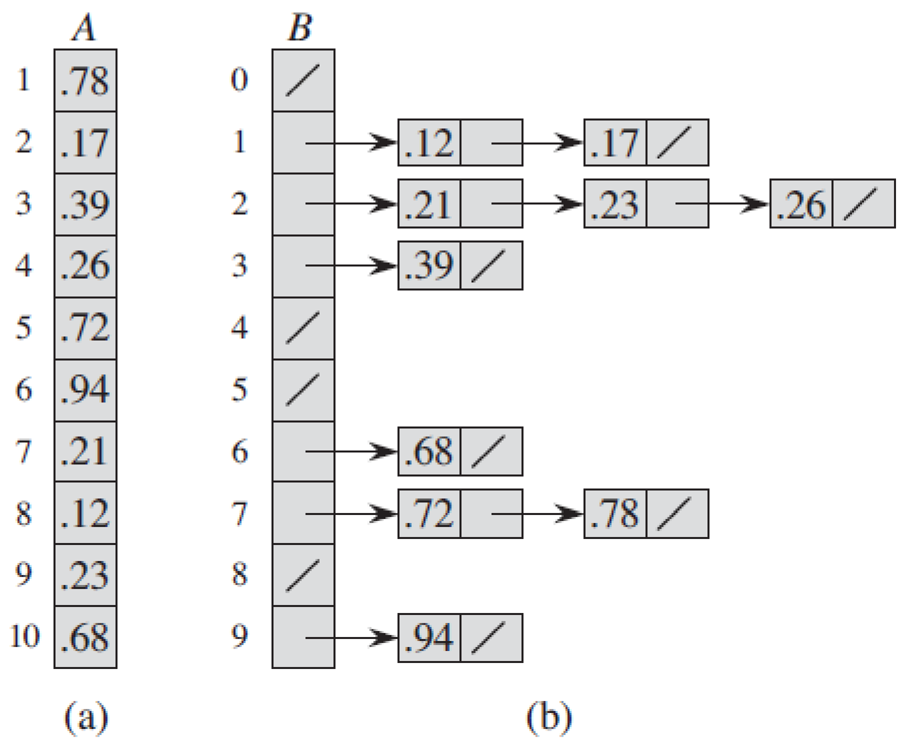

Figure 8.4 The operation of BUCKET-SORT for $n = 10$. **(a)** The input array $A[1..10]$. **(b)** The array $B[0..9]$ of sorted lists (buckets) after line 8 of the algorithm. Bucket i holds values in the half-open interval $[i/10, (i+1)/10)$. The sorted output consists of a concatenation in order of the lists $B[0], B[1], \ldots, B[9]$.

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BUCKET-SORT(A)
let B[0..n − 1] be a new array
n = A.length
for i = 0 to n − 1
    make B[i] an empty list
for i = 1 to n
    insert A[i] into list B[⌊nA[i]⌋]
for i = 0 to n − 1
    sort list B[i] with insertion sort
concatenate the lists B[0], B[1], ..., B[n − 1] together in order
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$$T(n) = \Theta(n) + \sum_{i=0}^{n-1} O(n_i^2) \; .$$

n_i denoting the number of elements placed in bucket $B[i]$.

$$\begin{aligned}
\mathrm{E}[T(n)] &= \mathrm{E}\left[\Theta(n) + \sum_{i=0}^{n-1} O(n_i^2)\right] \\
&= \Theta(n) + \sum_{i=0}^{n-1} \mathrm{E}\left[O(n_i^2)\right] \quad \text{(by linearity of expectation)} \\
&= \Theta(n) + \sum_{i=0}^{n-1} O\left(\mathrm{E}\left[n_i^2\right]\right) \quad \text{(by equation (C.22))} \; . \qquad (8.1)
\end{aligned}$$

$$\begin{aligned}
\mathrm{E}\left[n_i^2\right] &= \sum_{j=1}^{n} \frac{1}{n} + \sum_{1 \le j \le n} \sum_{\substack{1 \le k \le n \\ k \ne j}} \frac{1}{n^2} \\
&= n \cdot \frac{1}{n} + n(n-1) \cdot \frac{1}{n^2} \\
&= 1 + \frac{n-1}{n} \\
&= 2 - \frac{1}{n} \; ,
\end{aligned}$$

$$\mathrm{E}[T(n)] = \Theta(n) + n \cdot O(2 - 1/n) = \Theta(n).$$

$$X_{ij} = \mathrm{I}\{A[j] \text{ falls in bucket } i\}$$

$$n_i = \sum_{j=1}^{n} X_{ij} \ .$$

$$\begin{aligned} \mathrm{E}\left[n_i^2\right] &= \mathrm{E}\left[\left(\sum_{j=1}^{n} X_{ij}\right)^2\right] \\ &= \mathrm{E}\left[\sum_{j=1}^{n}\sum_{k=1}^{n} X_{ij}X_{ik}\right] \\ &= \mathrm{E}\left[\sum_{j=1}^{n} X_{ij}^2 + \sum_{1\le j\le n}\ \sum_{\substack{1\le k\le n \\ k\ne j}} X_{ij}X_{ik}\right] \\ &= \sum_{j=1}^{n} \mathrm{E}\left[X_{ij}^2\right] + \sum_{1\le j\le n}\ \sum_{\substack{1\le k\le n \\ k\ne j}} \mathrm{E}\left[X_{ij}X_{ik}\right] \ , \end{aligned}$$

$$\begin{aligned} \mathrm{E}\left[X_{ij}^2\right] &= 1^2 \cdot \frac{1}{n} + 0^2 \cdot \left(1 - \frac{1}{n}\right) \\ &= \frac{1}{n} \ . \end{aligned}$$

$$\begin{aligned} \mathrm{E}\left[X_{ij}X_{ik}\right] &= \mathrm{E}\left[X_{ij}\right]\mathrm{E}\left[X_{ik}\right] \\ &= \frac{1}{n} \cdot \frac{1}{n} \\ &= \frac{1}{n^2} \ . \end{aligned}$$

$$\mathrm{E}\left[n_i^2\right] = 2 - 1/n$$